

BLOODCHAIN

Confidential Codebase & Intellectual Property Protection Report

A detailed technical inventory of the Bloodchain platform, its product applications, shared services, authored workflows, evidence locations, and third-party boundaries.

DOCUMENT STATUS

Confidential — internal use only. Prepared for founder, investor and institutional pitch preparation. This report records the current workspace snapshot as of **19 August 2026**; it is not a legal opinion, patentability assessment, copyright registration, or substitute for advice from Botswana-qualified intellectual-property and data-protection counsel.

Prepared for	Purpose	Evidence basis
Bloodchain leadership	Preserve and communicate what has been created before external pitching	Current source tree, package manifests, API contract, database schema, artifact configuration and generated deliverables

Working position: Bloodchain is a Botswana-focused, multi-role blood-supply coordination platform. Its strongest authored assets are its domain model, operational workflows, laboratory state controls, chain-of-custody implementation, API contract, original product composition and Bloodchain brand expression—not the underlying open-source frameworks.

1. Purpose, scope and handling

This document is a technical and product-origin inventory for internal IP protection. It identifies the software, workflows, content, visual expression and evidence locations that together make up Bloodchain. It is designed to help the team describe its work consistently during investor, government and health-sector discussions; preserve a record of pre-disclosure development; and separate proprietary contribution from third-party software.

HOW TO USE THIS REPORT

Keep this PDF with its source repository export, Git history, dated build artifacts, contributor records, design files and signed confidentiality/assignment records. Before disclosing detailed architecture, share only the level of detail appropriate to the audience and use an NDA or institutional confidentiality arrangement where suitable.

Included

- All registered Bloodchain artifacts: public demo, web donor portal, mobile donor app, hospital terminal, national dashboard, laboratory workstation, API service, pitch deck and internal mockup sandbox.
- Shared API specification, React Query client, server-side Zod validators, database schema, seed/demo data and repository tooling.
- Product flows, endpoints, data entities, laboratory rules, evidence paths and an authored-versus-generated/third-party boundary.

Not included or not asserted

- Any personal secrets, credentials, private keys, database connection values or user data.
- Ownership of React, Vite, Expo, PostgreSQL, Drizzle, Zod, Orval, Tailwind, Radix/shadcn-derived primitives, Replit services, EAS or other third-party/open-source dependencies.
- A claim that the current demonstration is clinically certified, production authorised, integrated with laboratory instruments/EHRs, or compliant with every applicable health-data requirement.

2. Executive asset register

Asset / code package	Audience and purpose	Primary delivery surface	Authored value represented
Bloodchain Demo @workspace/demo	Investors, partners and public-facing narrative	Web root /	Brand story, national-system explanation, product launcher, presentation content
Pulse Web @workspace/pulse	Blood donors	Web /pulse/	Registration, donor dashboard, history, donor card, facilities and guidance
Pulse Mobile @workspace/pulse-mobile	Blood donors on mobile	Expo / Android build	Native donor registration/card workflow, persistent donor context, shareable card export
Sanctum @workspace/sanctum	Hospital blood-bank staff	Web /sanctum/	Facility inventory, unit logging, transfer routing and ledger views
Vigil @workspace/vigil	National/ministry operations	Web /vigil/	National telemetry, facility visibility, shortage alerts and routing tool
Crucible @workspace/crucible	Blood-processing laboratory teams	Web /crucible/	Screening, component separation, state transitions and chain-of-custody ledger
API Server @workspace/api-server	All Bloodchain applications	API /api	Shared business API, routing logic, laboratory guards and service orchestration
Bloodchain Pitch @workspace/bloodchain-pitch	Pitch meetings	Web /bloodchain-pitch/	Ten-slide narrative, presentation viewer and slide interaction model
Mockup Sandbox @workspace/mockup-sandbox	Internal design workflow	Internal web artifact	Component-preview infrastructure; supporting tooling rather than a deployed product surface

3. Platform architecture and original system design

Bloodchain is implemented as a pnpm/TypeScript monorepo. Multiple role-specific user interfaces consume a common API contract and a common PostgreSQL-backed operational model. The architecture keeps domain rules in the API/database layer while giving each user group a distinct, tailored surface. This is a material product-design decision: donor engagement, hospital operations, national coordination and laboratory processing are represented as connected, but separate, workflows.

Layer	Source of truth	What it contains	IP / evidence note
Product applications	artifacts/*	React/Vite web applications, Expo mobile application and authored slide deck	Custom pages, flows, narrative, branding and composition are the relevant authored expression
API contract	lib/api-spec/openapi.yaml	Public operations, request/response schemas and laboratory data types	Primary evidence of the platform interface and intended system behavior
Server implementation	artifacts/api-server/src	Express routes, facility/routing behavior, lab service and transactional mutations	Primary evidence for custom business logic
Data model	lib/db/src/schema	Donor, blood-unit, facility, laboratory processing, component and event tables	Primary evidence for the domain model and lifecycle
Shared client/validation	lib/api-client-react and lib/api-zod	Generated React Query hooks and generated Zod validators from OpenAPI	Generated outputs are derivative artifacts; retain source OpenAPI as the authored master
Deployment/tooling	artifact TOMLs, root manifests, scripts	Artifact routes, builds, workspace configuration and post-merge script	Operational evidence; do not confuse platform configuration with sole product IP

Role-aware product system

- **Donor side:** Pulse web and Pulse Mobile present registration, eligibility and personal donation history in donor-friendly language and visuals.
- **Facility side:** Sanctum gives hospital staff inventory, unit intake, inter-facility transfer and ledger views.
- **National side:** Vigil aggregates stock, shortage and facility information into a ministry/command-centre interface with refresh cadence and routing support.
- **Laboratory side:** Crucible establishes a dedicated processing workflow from test intake through screening, component separation, quarantine/release/discard, with a chronological hashed lab event trail.
- **Shared operational core:** the API and database bring those surfaces together around donors, facilities, blood units and laboratory processing records.

Authoring and generation boundary

The authored OpenAPI document drives code generation into the React client and Zod validator packages through Orval. Treat `lib/api-spec/openapi.yaml` and the authored server/database implementations as the stronger evidence of original work. Treat files under `src/generated`, compiled `dist` output and vendor packages as generated or third-party material unless separately modified and documented.

4. Bloodchain Demo — public ecosystem and partner narrative

Route / runtime	Audience	Product role
/	Investors, prospective partners and public-facing audiences	A single-page, editorial product narrative and launcher that frames the Bloodchain problem, architecture, roadmap and connected applications.

What this application contains

- Hero/mission sequence, live national-stat display, product architecture narrative and roadmap treatment.
- Expandable roadmap items that open embedded product experiences; mobile application download/link presentation.
- Custom ECG-style BLOODCHAIN mark, Botswana/ledger/architecture imagery, dark editorial visual system and Framer Motion reveal/accordion behavior.
- Founder/contact call-to-action, a local demo-access modal and local media/lightbox state.

Principal user flows

- Read current national statistics; explore system architecture and ledger story; inspect product roadmap; open an embedded product experience or mobile link; start a partner contact action.

Data/API relationship

Uses the national-stats API hook for live summary values. The remaining narrative, imagery and interaction structure are local presentation content.

Evidence locations

- artifacts/demo/src/App.tsx
- artifacts/demo/src/pages/home.tsx
- artifacts/demo/src/components/*
- artifacts/demo/src/index.css
- artifacts/demo/public/botswana-map.jpg
- artifacts/demo/public/ledger-abstract.jpg
- artifacts/demo/public/sovereign-architecture.jpg

IP PROTECTION POSITION

Claim the authored narrative, information architecture, product composition, bespoke graphical mark, custom interaction composition and original media where ownership/permission is documented. Do not claim generic animation libraries, React or stock/third-party assets without a verified license record.

5. Pulse Web — donor portal

Route / runtime	Audience	Product role
/pulse/	Blood donors	A web portal for donor discovery, registration, personal donor records, education and facility discovery.

What this application contains

- Routes for landing, registration, dashboard, knowledge, facilities and a not-found state.
- Clinical/profile registration form, donor-ID handoff, donor dashboard and donation-history filter.
- Browser donor-card download flow, eligibility/compatibility education, facility listing and persistent demo panel.
- Warm cream/red donor-facing visual identity distinct from operational applications.

Principal user flows

- View supply context; register a donor; carry the returned donor ID into the dashboard; retrieve donor profile/history; filter history; download a donor card; discover facilities and review donation guidance.

Data/API relationship

Consumes generated React Query hooks for national statistics, donor registration/retrieval, blood-unit history and facilities. It uses the shared OpenAPI client rather than duplicating server contracts.

Evidence locations

- artifacts/pulse/src/App.tsx
- artifacts/pulse/src/pages/home.tsx
- artifacts/pulse/src/pages/register.tsx
- artifacts/pulse/src/pages/dashboard.tsx
- artifacts/pulse/src/pages/knowledge.tsx
- artifacts/pulse/src/pages/facilities.tsx
- artifacts/pulse/src/components/*
- artifacts/pulse/src/index.css

IP PROTECTION POSITION

The claimable value is the donor workflow, custom page composition, educational framing, donor-card experience and Bloodchain-specific visual expression. The common UI primitives and React Query machinery are third-party/configured components.

6. Pulse Mobile — native donor companion

Route / runtime	Audience	Product role
Expo / Android build	Blood donors using mobile devices	A React Native/Expo donor experience that carries the Pulse flow into a native tabbed application.

What this application contains

- Home, Register and My Card/Dashboard tab experiences, plus root layout and not-found handling.
- Persistent donor-ID context backed by AsyncStorage; refresh and clear/start-over paths.
- Native sharing/ViewShot donor card export and a custom cross-browser PNG download implementation for web use.
- Platform-specific tab presentation: iOS NativeTabs liquid-glass path and Android/web blur treatment.

Principal user flows

- View national stock context; register with donor details; retain donor identity locally; load personal profile and owned units; share or download the donor card; refresh or reset the local session.

Data/API relationship

Uses the same shared API operations as Pulse Web: national stats, donor registration, donor lookup and donor-filtered units. App configuration sets the mobile API origin separately from same-origin web clients.

Evidence locations

- artifacts/pulse-mobile/app/_layout.tsx
- artifacts/pulse-mobile/app/(tabs)/_layout.tsx
- artifacts/pulse-mobile/app/(tabs)/index.tsx
- artifacts/pulse-mobile/app/(tabs)/register.tsx
- artifacts/pulse-mobile/app/(tabs)/dashboard.tsx
- artifacts/pulse-mobile/lib/donor-context.tsx
- artifacts/pulse-mobile/lib/donor-card-download.ts
- artifacts/pulse-mobile/app.json

IP PROTECTION POSITION

The original work includes the mobile flow, persistence pattern, donor card/export experience and platform-specific presentation choices. Expo/React Native, native modules and EAS are third-party/platform infrastructure.

7. Sanctum — hospital blood-bank workstation

Route / runtime	Audience	Product role
/sanctum/	Hospital blood-bank technicians and facility operations staff	A facility-focused terminal for tracking local inventory, receiving/logging units, viewing ledger information and requesting supply routing.

What this application contains

- Routes for overview, log unit, transfers, ledger and not-found.
- Selected-facility context, inventory and national-status display, facility-unit intake form and local ledger filtering.
- Shortage and facility discovery view with a request-routing action.
- Dark scanline/command-console visual system, facility badge and blue-green operational language.

Principal user flows

- Select a facility; inspect inventory/national context; log a blood unit; identify shortage/facility availability; submit a route request; view the ledger-filtered unit list.

Data/API relationship

Consumes facility inventory, national statistics, ledger summary, facility discovery, blood-unit logging, national-shortage and route-request operations from the shared client.

Evidence locations

- artifacts/sanctum/src/App.tsx
- artifacts/sanctum/src/components/layout.tsx
- artifacts/sanctum/src/pages/overview.tsx
- artifacts/sanctum/src/pages/log-unit.tsx
- artifacts/sanctum/src/pages/transfers.tsx
- artifacts/sanctum/src/pages/ledger.tsx
- artifacts/sanctum/src/index.css

IP PROTECTION POSITION

The claimable contribution is the facility workflow, custom terminal composition, local inventory/transfer experience and Bloodchain-specific interaction design. It is not a claim to generic dashboard widgets or underlying UI libraries.

8. Vigil — national command dashboard

Route / runtime	Audience	Product role
/vigil/	National/ministry coordinators and operational oversight teams	A national-level command view that turns facility, inventory and shortage data into operational awareness and supply-routing support.

What this application contains

- Routes for overview, facility visibility, alerts/routing and not-found.
- National inventory/shortage/ledger panels with distinct periodic refresh intervals.
- Facility search and expandable facility inventory; severity-coded shortage alerts; route calculator.
- Ministry-scale dark telemetry style with charts, live clock and operational panels.

Principal user flows

- Monitor national stock and deficit signals; inspect a facility; search/expand inventory; review alerts; create a compatible route recommendation for a blood request.

Data/API relationship

Consumes national stats, national shortage, ledger summary, facilities, facility inventory and route-request operations. The client refresh cadence is implemented in the Vigil page layer.

Evidence locations

- artifacts/vigil/src/App.tsx
- artifacts/vigil/src/components/layout.tsx
- artifacts/vigil/src/pages/Overview.tsx
- artifacts/vigil/src/pages/Facilities.tsx
- artifacts/vigil/src/pages/Alerts.tsx
- artifacts/vigil/src/index.css

IP PROTECTION POSITION

The claimable value lies in the national coordination product composition, operational telemetry framing, facility/shortage workflow and Bloodchain-specific decision presentation. Underlying chart/UI libraries must remain attributed to their respective licenses.

9. Crucible — blood-processing laboratory workstation

Route / runtime	Audience	Product role
/crucible/	Blood-processing laboratory personnel	A dedicated laboratory application that turns collected units into screened and separated components while enforcing a defined processing state model and event trail.

What this application contains

- Dashboard, searchable/filterable worklist, unit detail and laboratory ledger routes.
- Screening record for ABO/Rh and HIV, Hepatitis B/C, malaria and syphilis markers.
- Component separation for red cells, plasma and platelets; processing/quarantine/release/discard actions.
- Stage and risk badges, terminal/expired read-only behavior, mutation feedback and event-history display.
- Dense dark laboratory-instrument visual language with monospace identifiers and high-contrast status coding.

Principal user flows

- Open the worklist; filter by stage/risk; inspect a unit; record screening; move an eligible unit into processing; create components; quarantine/release/discard subject to rules; review chronological audit events and chain hashes.

Data/API relationship

Consumes the dedicated laboratory API: dashboard, worklist, unit detail, screening submission, component separation, transition and event stream. Client-side mutations invalidate/reload dashboard, worklist, detail and ledger data.

Evidence locations

- artifacts/crucible/src/App.tsx
- artifacts/crucible/src/components/layout.tsx
- artifacts/crucible/src/components/badges.tsx
- artifacts/crucible/src/pages/dashboard.tsx
- artifacts/crucible/src/pages/worklist.tsx
- artifacts/crucible/src/pages/unit-detail.tsx
- artifacts/crucible/src/pages/ledger.tsx
- artifacts/crucible/src/index.css
- screenshots/crucible-dashboard.jpg

IP PROTECTION POSITION

This is one of the strongest standalone product assets: its laboratory workflow, screening/component model, state-guarded transitions, event-chain presentation and distinct workstation expression are concrete original implementation areas. It should be protected with careful contributor assignment and source-history preservation.

10. Bloodchain Pitch — interactive slide deck

Route / runtime	Audience	Product role
/bloodchain-pitch/	Investors, partners and presentation audiences	A ten-slide pitch artifact with authored narrative, fixed-canvas slide components and a browser presentation viewer.

What this application contains

- Ten individual slide components, a slide manifest, contact-sheet route and viewer route.
- Keyboard, click, touch and parent-frame navigation; a fixed 1920×1080 design contract adapted to responsive viewing.
- Slides covering problem framing, system narrative, partnership and build call-to-action.
- Slide data/position tooling for editor validation and repair.

Principal user flows

- Start the viewer; advance/reverse slides using keyboard, click or touch; open individual slides or all-slides contact sheet; allow a host frame to control navigation through messages.

Data/API relationship

It has no operational API dependency. The narrative, slide layout/components, viewer behavior and presentation styling are authored content.

Evidence locations

- artifacts/bloodchain-pitch/src/App.tsx
- artifacts/bloodchain-pitch/src/slideLoader.ts
- artifacts/bloodchain-pitch/src/pages/slides/Slide1.tsx through Slide10.tsx
- artifacts/bloodchain-pitch/src/data/slides-manifest.json
- artifacts/bloodchain-pitch/src/index.css
- artifacts/bloodchain-pitch/scripts/*

IP PROTECTION POSITION

Protect the original storyline, visual composition, slide-specific copy and interactive viewer composition. The presentation framework and any third-party image/fonts require their own attribution or rights review.

11. Mockup Sandbox — internal component preview environment

Route / runtime	Audience	Product role
Internal artifact / component preview server	Bloodchain design and implementation workflow	A development-only component-preview surface used to render, compare and iterate on UI components without making it part of the end-user Bloodchain product.

What this application contains

- Vite-based preview shell, mockup preview plugin and source application/styles.
- Artifact registration as a design/canvas workspace and a dedicated preview-server workflow.

Principal user flows

- Host isolated component previews during design iteration; use as supporting evidence of the team's design/development process rather than as a public product module.

Data/API relationship

This artifact does not represent a business-user workflow. It supports the production applications but should be categorised separately in commercial and IP discussions.

Evidence locations

- artifacts/mockup-sandbox/src/App.tsx
- artifacts/mockup-sandbox/src/index.css
- artifacts/mockup-sandbox/mockupPreviewPlugin.ts
- artifacts/mockup-sandbox/.replit-artifact/artifact.toml

IP PROTECTION POSITION

Claim only any original preview orchestration or components actually authored here. Do not overstate it as a client-facing Bloodchain module; it is internal tooling.

12. Shared API, domain contract and database

The shared back end is the operational backbone for every product surface. It is an Express service backed by Drizzle/PostgreSQL. The HTTP interface is declared once in an OpenAPI specification, then used to generate typed React Query hooks and Zod validators. This structure is valuable both technically and as evidence: it makes the relationship between domain rules, contract and clients traceable.

API operation inventory

Domain	Operations under the API root	Implemented purpose
Health	GET /health	Service health/status response.
Donors	GET /donors; POST /donors; GET /donors/{donorId}	List/filter donors, register a donor, retrieve a donor record.
Blood units	GET /units; POST /units; GET /units/{unitId}	List/filter units, log a unit and retrieve a unit.
Facilities	GET /facilities; GET /facilities/{facilityId}; GET /facilities/{facilityId}/inventory	Find facility records and inspect local inventory/critical categories.
Routing	POST /route; GET /route/national-shortage	Rank available supply candidates and summarise shortage against threshold data.
National statistics	GET /stats/national; GET /stats/ledger	Aggregate national operational metrics and a presentation-oriented ledger summary.
Laboratory	GET /lab/worklist; GET /lab/units/{unitId}; POST /lab/units/{unitId}/screening; POST /lab/units/{unitId}/components; POST /lab/units/{unitId}/transition; GET /lab/dashboard; GET /lab/events	Serve and mutate the dedicated processing worklist, laboratory unit detail, component records, state transitions and event history.

Core data entities

Entity	Key responsibility	Selected content / relationship
donors	Donor identity and eligibility context	Donor ID, contact/profile information, blood type, district, totals, status and eligibility dates.
blood_units	Collected unit lifecycle	Unit ID, donor/facility references, blood type, status, collection/expiry time, temperature and a base chain hash.
facilities	Networked storage/threshold context	Facility identity, district, type/status, JSON inventory and threshold values.
lab_processing	One laboratory processing record per unit	Facility, stage, risk, screening JSON, event sequence, chain head and timestamps.
lab_components	Separated component records	Component identity, parent unit, type, volume, status and expiry.
lab_events	Chronological laboratory event trail	Event ID, unit/facility, action, actor, reason, sequence, timestamp and hash; indexed for unit/facility/time retrieval.

Evidence locations

- lib/api-spec/openapi.yaml — authored contract and schema source.
- artifacts/api-server/src/routes/index.ts and route modules — endpoint implementation boundary.
- lib/db/src/schema/donors.ts, blood_units.ts, facilities.ts, laboratory.ts — persistent domain model.
- lib/db/src/seed.ts, demo-data.ts, lab-demo-data.ts — deterministic demonstration dataset and operational scenarios.
- lib/api-spec/orval.config.ts — code generation configuration and source/output relationship.

13. Laboratory workflow and chain-of-custody mechanics

Crucible's server-side laboratory rules are a focused source of original implementation. The workflow does not simply render statuses: it uses transactional mutation logic and row locking to protect unit processing. The following description records implemented behavior, not a claim of clinical certification.

Lifecycle stage	Implemented transition / guard	Recorded event behavior
awaiting_tests	Initial test queue; may move to processing when screening outcome is suitable.	Event appended with actor, reason/time context and next sequence.
processing	Used for clear screened unit processing; component separation requires completed clear screening.	Screening and processing actions are retained in the unit event stream.
quarantine	Used for reactive/review results and separated components; release requires cleared screening, components and unexpired products.	Quarantine requires a reason in the state-transition input.
released	Terminal release state after prerequisites are met; later transitions/re-screening are rejected.	Release event completes the demonstrable processing trail.
discarded	Terminal state; requires a reason and synchronises relevant status treatment.	Discard event records reason/actor context.

Screening, separation and risk derivation

- Screening input includes ABO group, Rh factor, HIV, Hepatitis B, Hepatitis C, malaria and syphilis markers.
- Risk is derived from results: positive markers become reactive; pending results become pending; ABO/Rh mismatch becomes review; otherwise the outcome can be clear.
- Component separation validates an eligible clear-processing unit, prevents duplicate component types, limits declared volume and creates red-cell, plasma and platelet component records with distinct shelf-life assumptions.
- Release is blocked unless the implemented prerequisites are satisfied; expired base units or components and terminal/transfused states are rejected.

Event-chain implementation

Each laboratory event is appended in a transaction after the processing record's prior sequence and chain head are read. The next hash is SHA-256-derived from the prior hash plus the unit, facility, sequence, action, actor, reason and timestamp context; the processing record is updated with the new sequence/head. Startup logic also repairs legacy entries missing explicit sequence/head data. This gives the product a concrete chain-of-custody design to preserve and mature, while not by itself constituting an externally validated blockchain or legal non-repudiation system.

IMPORTANT CLAIM BOUNDARY

The current national /stats/ledger summary is a presentation-oriented ledger statistic and is distinct from the laboratory lab_events chain. When pitching, describe the laboratory event chain precisely; do not imply every national metric is backed by the same immutable event mechanism.

Key evidence locations

- artifacts/api-server/src/lib/lab-mutations.ts — transactional guards, state transitions, component logic and event append.
- artifacts/api-server/src/lib/lab-service.ts — worklist/detail/metric/event read service and risk derivation.
- lib/db/src/lab-ledger.ts — compatibility repair for legacy event sequence/chain head.
- lib/db/src/schema/laboratory.ts — processing/component/event tables and indexes.

14. Shared libraries, build system and operating model

Package / location	Role	Ownership and evidence treatment
lib/api-spec	OpenAPI source document and Orval generation configuration.	Treat the authored OpenAPI document and its Bloodchain domain schema as key source evidence.
lib/api-client-react	Generated React Query operations plus custom fetch wrapper.	Generated portions are derivative from the spec; custom wrapper is authored support code.
lib/api-zod	Generated server request/response validators and type exports.	Generated portions are derivative from the spec; preserve for build reproducibility, but anchor claims in the OpenAPI source.
lib/db	Drizzle connection, schema, sequence helpers, seed/demo datasets, laboratory data and ledger repair.	High-value authored domain implementation.
artifacts/api-server	Express application, route mounting, logging, lab service/mutations and startup seed/init sequence.	High-value authored service implementation.
pnpm root manifests / artifact TOMLs / scripts	Workspace, build, artifact routing, production metadata and post-merge helper.	Operational evidence; platform/tool configuration itself has narrower protectable scope.

Runtime and deployment characteristics

- Private pnpm workspace with Node.js/TypeScript; root build performs library typechecking and package builds.
- Web artifacts use Vite with injected PORT and BASE_PATH, static output and SPA-style route handling.
- The API is Express/Drizzle/PostgreSQL and requires a database connection; startup runs idempotent facility/demo/laboratory initialisation.
- The mobile application is Expo/React Native; EAS profiles support an internal Android APK and a production Android App Bundle.
- Replit artifact metadata registers the delivery surfaces. This is deployment infrastructure, not a substitute for a formal source-control, CI/CD and release-evidence policy.

Third-party and generated boundary

Category	Examples in the repository	Recommended treatment in an IP pack
Open-source application stack	React, React DOM, Vite, TypeScript, Wouter, TanStack React Query, Tailwind, Frammer Motion, Lucide, Radix/shadcn-style primitives.	Retain license notices and package lockfile. Do not claim ownership; claim the Bloodchain-specific orchestration, workflows and presentation.
API/data stack	Express, PostgreSQL/node-postgres, Drizzle ORM, Zod, drizzle-zod, Orval, esbuild, Pino.	Treat as dependencies. Preserve version/lock evidence and distinguish authored route/schema/mutation code.
Mobile/platform stack	Expo, Expo Router, React Native, EAS and related native modules.	Treat as third-party platform tooling. Claim native workflow composition and authored application code, not the platform.
Replit platform tooling	Artifact metadata, Replit Vite plugins, connectors SDK, managed workflow/deployment facilities.	Treat as platform infrastructure; preserve configurations as operational evidence, not as exclusive technology ownership.
Generated output	lib/api-client-react/src/generated, lib/api-zod/src/generated, compiled dist output.	Preserve for reproducibility but cite OpenAPI and authored source code as primary evidence.

15. IP claim map and evidence to preserve

Potential proprietary asset	Concrete implementation evidence	Protection / preservation action
Bloodchain product architecture	Multi-role donor, facility, ministry and lab surfaces joined through a shared API/domain model.	Keep versioned architecture diagrams, Git history, pitch chronology and dated releases. Describe as a product/system design, not a claim to generic health dashboards.
Donor engagement workflows	Pulse Web/Mobile registration, donor identity handoff, history, card export, facilities and education.	Preserve source, UX flows, visual comps and records of authorship. Review wording/imagery licenses.
Facility allocation and shortage workflows	Facility inventory/threshold data, national shortage computation, compatible candidate routing and user interfaces.	Preserve route/routing source, API contract and scenario test evidence. Avoid claiming unverified clinical optimisation performance.
Laboratory lifecycle rules	Screening results, risk derivation, component separation, quarantine/release/discard constraints and terminal protections.	Preserve lab mutation code, schema, UI and test records. Seek clinical/legal review before production claims.

Potential proprietary asset	Concrete implementation evidence	Protection / preservation action
Chain-of-custody event mechanism	Per-unit sequence, hash head, hash payload, transactional append and ordered ledger display.	Preserve source history and example event exports; explain its implemented scope accurately.
Brand and creative expression	Bloodchain/Crucible/Sanctum/Vigil/Pulse names, visual language, slides, logo mark, original copy and custom page composition.	Run trademark/name clearance; keep original source/design files; record ownership or licence of every image/font.
OpenAPI/domain vocabulary	Explicit endpoints and schema names for donors, units, facilities, routing and laboratory processing.	Archive versioned spec snapshots and generated output mapping; use contributor agreements for future API changes.

16. Security, data-sharing and production-readiness boundary

This report is about technical authorship and evidence, but safe external pitching requires clear statements about what is demonstrable today and what still needs institutional governance before real data or clinical operations are introduced.

Current observed state	Implication for a partner discussion	Recommended response before/alongside a pilot
Demo/seeded operational records are present.	Do not represent the displayed records as live NBTS clinical or national data.	Use labelled synthetic/demo datasets; document source/permission for all imagery and scenario data.
Crucible has no production authentication or role enforcement in the current implementation; operator name is currently supplied by callers.	Do not present release/discard actions as ready for operational deployment.	Add authenticated identities, facility-scoped RBAC, server-derived actor attribution and audit review before real use.
Laboratory hardware, EHR/LIS and courier integrations are out of current scope.	Do not imply automated instrument ingestion or end-to-end clinical interoperability.	Define interface, verification, exception and accountability requirements with NBTS.
Database schema is deployed via an automated push-style build command.	Change governance and production migrations require more formal control before institutional rollout.	Adopt migration review, backup/rollback and change-approval practices.
The codebase contains third-party packages and generated material.	Ownership claims need an open-source/commercial-asset boundary.	Create an SBOM/licence register; retain lockfile and notices; complete image/font provenance review.

17. Evidence preservation checklist before pitching

- **Freeze a dated source snapshot.** Create a signed/tagged Git commit, archive it in a controlled repository and retain this report alongside the exact commit hash.
- **Record authorship and assignment.** Keep contributor identity, employment/contractor IP assignment, work dates and review records for code, design, copy, images and decks.
- **Hash the evidence package.** Generate checksums for source export, PDF report, pitch deck export and key design assets; store the signed/checksummed register separately.
- **Preserve build evidence.** Archive build logs, production/dev artifact configuration, package lockfile and released application captures.
- **Clear names and creative assets.** Obtain trademark/name advice for Bloodchain and product names; record the provenance or licence for logo elements, photos, fonts and illustrations.
- **Control external disclosure.** Use a redacted capability deck for early conversations. Move detailed API/schema/lab-chain explanations into governed technical sessions under suitable confidentiality terms.
- **Prepare data-governance material.** Before receiving any identifiable health data, agree a written data-sharing framework covering purpose, legal basis, minimum dataset, security, retention, breach handling, ownership, access and deletion/return.

LEGAL NOTE

This inventory is evidence-oriented, not a legal registration. Engage qualified Botswana IP, commercial and data-protection counsel to advise on trademarks, copyright assignment, database rights, confidentiality, procurement, health-data governance and any pilot agreement before relying on this report in a legal dispute or regulated deployment.

Appendix A — source evidence index

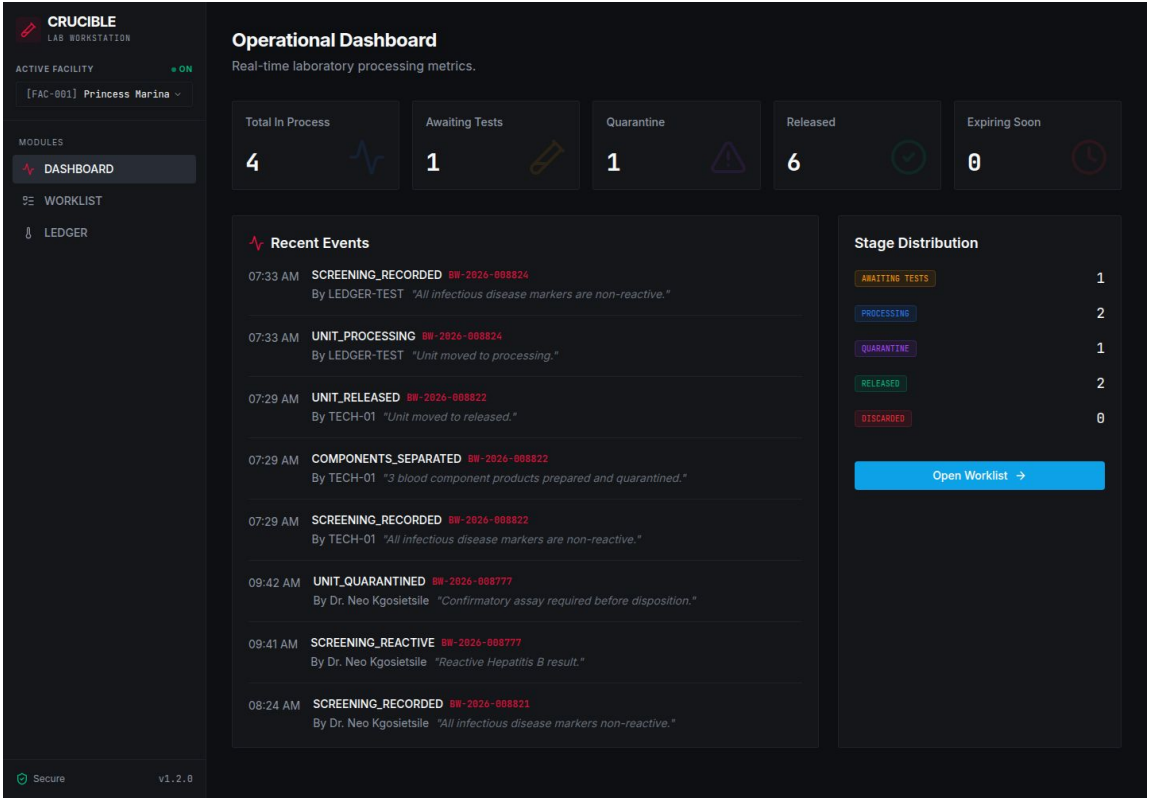
The following index lists the principal authored locations used in this report. It is intentionally a source map rather than a full file dump; the complete repository and Git history remain the authoritative granular evidence.

Area	Principal source locations	What the source evidences
Public narrative	artifacts/demo/src; artifacts/demo/public; artifacts/demo/.replit-artifact/artifact.toml	Public product narrative, original page composition/assets, deployment surface.
Donor web	artifacts/pulse/src/App.tsx; src/pages; src/components; src/index.css	Registration/dashboard/knowledge/facility/card experience.

Area	Principal source locations	What the source evidences
Donor mobile	artifacts/pulse-mobile/app; lib/donor-context.tsx; lib/donor-card-download.ts; app.json	Native flows, local donor persistence and card export.
Hospital operations	artifacts/sanctum/src/App.tsx; components/layout.tsx; pages; index.css	Facility inventory, unit intake, transfer and facility ledger surface.
National operations	artifacts/vigil/src/App.tsx; components/layout.tsx; pages; index.css	National dashboard, shortage, facilities and routing surface.
Laboratory operations	artifacts/crucible/src; artifacts/api-server/src/lib/lab-*; lib/db/src/schema/laboratory.ts	Screening, processing, component, state and chain-of-custody experience.
Shared contract	lib/api-spec/openapi.yaml; lib/api-spec/orval.config.ts	Domain interface, request/response types and generation boundary.
Shared data	lib/db/src/schema; seed.ts; demo-data.ts; lab-demo-data.ts; lab-ledger.ts; sequences.ts	Persistent data model, demo datasets, ID/event support.
Server routes	artifacts/api-server/src/app.ts; index.ts; routes/*; lib/*	Express application, operational API routes, routing and laboratory business logic.
Pitch content	artifacts/bloodchain-pitch/src/App.tsx; pages/slides/*; slideLoader.ts; data/slides-manifest.json; index.css	Authored pitch narrative, interaction and slide composition.
Build/platform	package.json; pnpm-workspace.yaml; artifact TOMLs; scripts/post-merge.sh; replit.md	Workspace/build/deployment context and dependency policy.

Appendix B — visual evidence

Current Crucible operational dashboard capture from the running artifact. This is visual evidence of the authored laboratory dashboard and its populated workflow/event presentation; it is not evidence of a live clinical deployment.



End of confidential internal report